*Islamic University of Technology*

# Lab 5: Selection Statements

CSE 4108 : Structured Programming Lab

**Section 1B**

MIA, NFS, IAC, ASH

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# 1 Lab Tasks

## 1.1 Five For All

Write a C program that uses takes input an integer from the user and then prints whether the integer is divisible by 5 or not.

**[Sample Program]**
*Input 1*

```
Enter an integer: 3
```

*Output 1*

```
3 is not divisible
```

*Input 2*

```
Enter an integer: 45
```

*Output 2*

```
45 is divisible
```

## 1.2 Meter Converter

Write a C program that would take a value in meters(cm) as input from the user. If the input value is greater than or equal to 1000, the input is converted it to kilometers (km). If the input is between 1000 and 1, it is converted to centimeters (cm), or if it is less than or equal to 1, it is converted to millimeters (mm).

**[Sample Program]**
*Input 1*

```
Enter value in meters : 105.0
```

*Output 1*

```
105.00 m is 10500.00 cm
```

*Input 2*

```
Enter value in meters : 0.88
```

*Output 2*

```
0.88 m is 880.00 mm
```

**[Hint]** *Remember, output values are printed to two digits after decimal point.*

## 1.3 Auntie Virus

Your neighbourhood auntie has developed a software to scan and remove virus from computers. She needs your help to calculate how much to charge each user for this service. Each package has some flat price. Along with that, a percentage of the flat price is added as internet security charge. On top of that, **if the user takes mobile protection, another 300 TK is added with the overall price**. So, the equation for overall price is `flat price + internet security charge percentage of flat charge([OPTIONAL] + 300 TK if mobile protection)`.

Write a C program that would take the **duration** for which the user wants to use the product as input and whether the user wants mobile protection or not `(1/0)` as input from the user and print out the final charge of the product as output based on the following table.

| Package | Duration (month) | Flat Price (BDT) | Internet Security charge |
|---|---|---|---|
| Trial | 0-1 | 0.00 | 0% |
| AV | 2-6 | 200.27 | 10% |
| AV Plus | 7-12 | 500.50 | 15% |
| AV Premium | >12 | 1000.68 | 15% |

[*Extra 300 BDT is added for Mobile Protection]

Table 1: Auntie Virus Charging Scheme

**[Sample Program]**
*Input*

```
Enter Duration (Months): 8
Mobile Protection? (0/1): 1
```

*Output*

```
Overall Price (BDT): 875.58
```

**[Hint]** *Currencies are printed up to two places after decimal point.*

## 1.4 Train Time

Write a C program that will take as input the time when a train started its journey in 24hr format (00:00 to 23:59). The program will output the time when the train will end its journey, you can assume it takes 2 hours for the train to reach its destination.

**[Sample Program]**
*Input 1*

```
Enter journey start time: 00:43
```

*Output 1*

```
Journey end time: 02:43
```

*Input 2*

```
Enter journey start time: 23:16
```

*Output 2*

```
Journey end time: 01:16
```

**[Hint]** *Be careful to display output according to the range of the 24hr format (00:00 to 23:59).*

## 1.5 Changing Seasons

Write a C program that would take a date of 2021 in `dd-mm` format (*e.g. 27-01*) and print out the name of the bengali season that date belongs to. The following table shows each of the bengali seasons and their corresponding starting date and ending dates in `dd-mm` format.

| Bengali Season | Gregorian Calendar Date Range |
|---|---|
| Grishsho | 14-04 to 15-06 |
| Borsha | 16-06 to 17-08 |
| Shorot | 18-08 to 18-10 |
| Hemonto | 19-10 to 16-12 |
| Sheet | 17-12 to 12-02 |
| Boshonto | 13-02 to 13-04 |

Table 2: Bengali to Gregorian Calender Conversion for 2021

**[Sample Program]**
*Input*

```
Enter date (dd-mm) : 12-03
```

*Output*

```
Season: Boshonto
```

## 1.6 Grader

Write a C program that would take marks of a student as input and print out his/her grade and grade point based on table 3. **You can only use** `switch statement`. `if...else` statements are **not** allowed. Marks will be between 0 to 100.

| Grade | Marks | GP |
|---|---|---|
| A | 80 and Above | 4.00 |
| B | 70 to 79 | 3.50 |
| C | 60 to 69 | 3.00 |
| D | 50 to 59 | 2.50 |
| F | Below 50 | 0.00 |

Table 3: Reference Grade Sheet

**[Sample Program]**
*Input*

```
Enter marks: 75
```

*Output*

```
Grade: B
Grade Point: 3.50
```

**[Hint]** *Use some arithmetic calculations to cut down the number of* `switch cases` *required drastically.*